Physical Cumulative Exam
November 21, 2008

1. (a) What is the entropy change when 2 L of hot (65 °C) water is mixed with 1 L of cold (5 °C) water. For simplicity, assume that the heat capacity of water is constant C_p = 4.184 kJ/(kg K), and its density is 1 kg/L.

   (b) If the temperature outside is 10 °C and inside 25 °C, how much electrical power is needed to operate a heat pump which delivers 10 kW of heating power inside the house if the efficiency of the pump is 100 %?

   The absolute zero is 0 K = -273.2 °C. (Help: a heat pump is a heat engine running in reverse, i.e. it consumes work and transports heat from a colder to a hotter heat reservoir. The absolute values of the heat and work are the same as for a heat engine but the signs are reversed).